

Advanced Distributed Systems Examination

1. Consider a distributed consensus algorithm executing across 5 asynchronous nodes. Under the FLP impossibility result, what guarantee can be provided in an asynchronous network subject to crash failures?
- (A) Absolute liveness and safety simultaneously under all conditions
 - (B) Deterministic consensus within bounded time $O(n)$

- (C) Safety is preserved, but termination/liveness cannot be guaranteed deterministically
- (D) Fault tolerance up to $n/2$ Byzantine nodes without cryptographic signatures

2. Which consistency model requires that all operations appear to take effect at a specific point in time?

- (A) Eventual Consistency
- (B) Causal Consistency
- (C) Linearizability
- (D) Read-After-Write Consistency

=== ANSWER KEY ===

- 1. C
- 2. C